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BLOOMBERG SCHOOL
of PUBLIC HEALTH

Role of Epidemiology in Public Health

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Learning Objectives

- ▶ Describe the history of public health and epidemiology as scientific disciplines
- ▶ Explain the 10 essential public health services
- ▶ Understand the public health governance structure in the US
- ▶ Use a competency assessment tool to assess the abilities of components of the public health workforce to achieve the essential public health services



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Definition of Epidemiology

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Public Health Defined



- ▶ “The science and art of preventing disease, prolonging life, and promoting health through the organized efforts and informed choices of society, organizations, public and private communities, and individuals.”

—C-E. A. Winslow

- ▶ “Public health aims to provide maximum benefit for the largest number of people.”

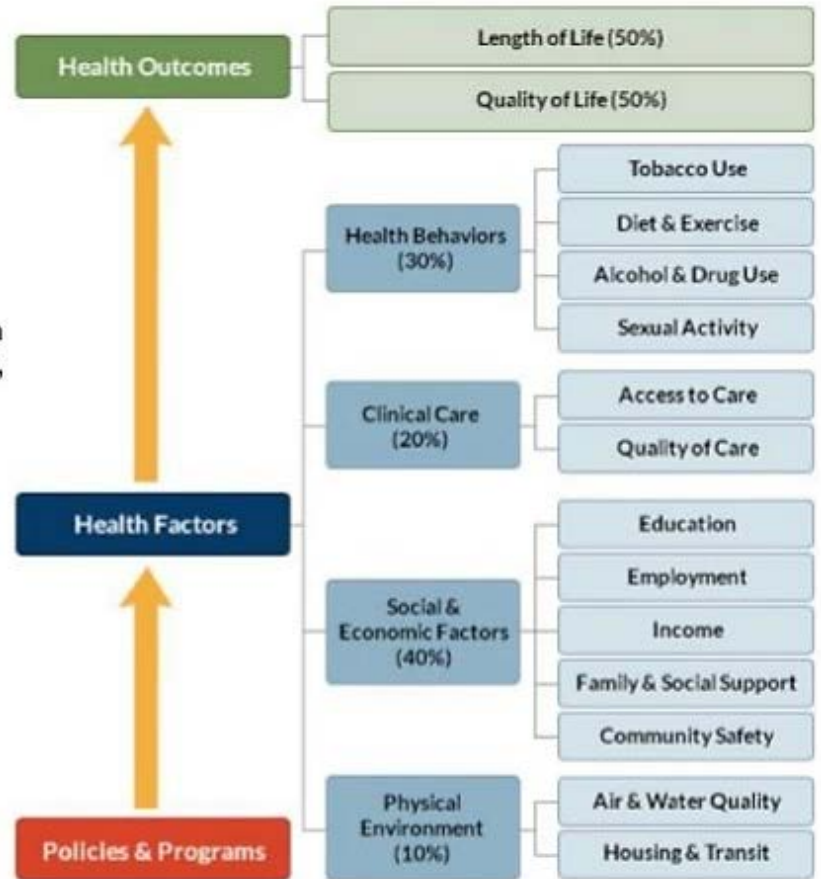
—World Health Organization

Framework: Management of Population Health

True Population Health Management

Requires a collaborative
strategy between leaders in
healthcare, politics, charity,
education, and business

Robert Wood
Johnson
Foundation,
2014



Epidemiology

- ▶ *Epi: on or upon; Demos: people; Logos: the study of*
- ▶ Epidemiology is the **study** of the **distribution** and **determinants** of **health-related states or events** in **specified populations**, and the **application** of this study to the control of health problems

Epidemiology Is a Core Component of Public Health

- ▶ Preview: exercise for this course
- ▶ Public Health Foundation: Core Competencies for Public Health Professionals
 - ▶ “A set of skills desirable for the broad practice of public health that professionals may want to possess as they work to protect and improve the nation’s health.”
 - ▶ “Offer a starting point for public health professionals in identifying professional development needs and developing training plans.”
- ▶ Assessment tool: divided into three tiers (by level of responsibility)
 - ▶ Tier 1: frontline/entry-level staff
 - ▶ Tier 2: program management/supervisory staff
 - ▶ Tier 3: senior management/executive-level staff

Where Are Skills in Epidemiology Needed?

- ▶ Tier 1:
 - ▶ Describe how public health sciences (including epidemiology) are used in public health delivery
 - ▶ Identify data that can be used for assessing the health of a community
 - ▶ Contribute to public health evidence base
- ▶ Tier 2:
 - ▶ Analyze validity, reliability, and comparability of data
 - ▶ Assess community health status and factors influencing health in a community
- ▶ Tier 3:
 - ▶ Ensure ethical principles are applied in accessing, collecting, analyzing, using, maintaining, and disseminating data
 - ▶ Resolve gaps in data
 - ▶ Determine trends from qualitative and quantitative data



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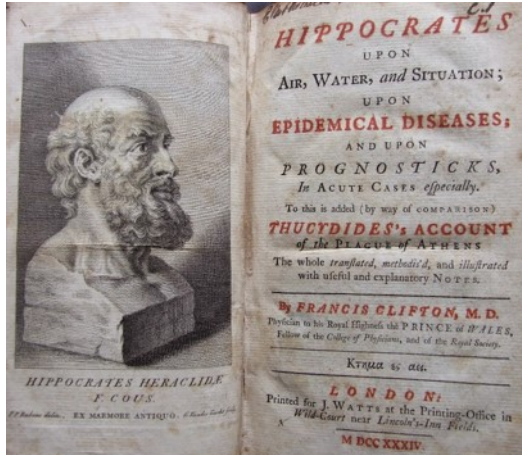
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History of Epidemiology

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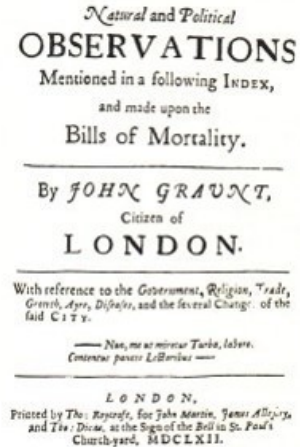
A Brief History of Epidemiology

400 BCE



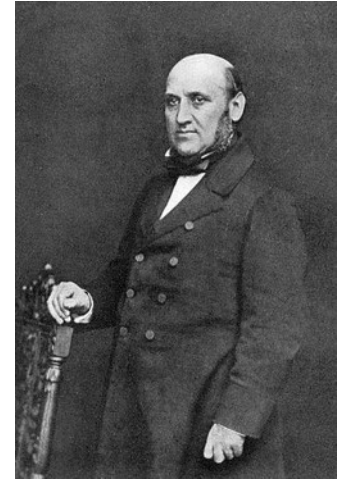
Hippocrates' *On air, water and places* explains how the environment and host factors play a role in disease development

1662



John Graunt: patterns of mortality by sex, urbanicity, season, etc.

1840



William Farr: first “vital statistics” tracking deaths by cause

London, Mid-1800s—1

- ▶ Sewage (as well as garbage, animal corpses, etc.) was thrown directly into London's Thames River, and unfiltered drinking water flowed from there to pumps throughout the city
- ▶ “The Great Stink” (July/August 1858): Parliament had to dismiss an ongoing meeting because the stench from the Thames was so strong



The Silent Highwayman (1858)



A drop of Thames water (1850)

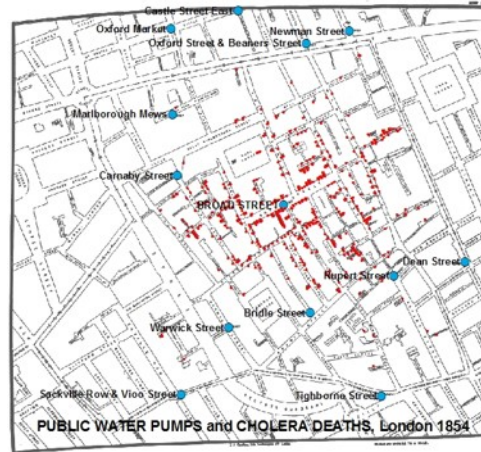
Cholera: A Major Cause of Mortality in Mid-1800s

- ▶ Cholera: diarrhea, often with nausea and vomiting
- ▶ Up to 50,000 estimated deaths per season (~20 per 1,000 population)
- ▶ Current theory: miasmas (air pollution) caused the illness
 - ▶ Areas closer to the river often had the worst air quality
- ▶ FYI—now known to be caused by the bacterium *Vibrio cholera* spread by contaminated water



“Boots on the Ground”

- ▶ Summer 1854: cholera was back with a vengeance
- ▶ Snow was convinced an intestinal germ was responsible
- ▶ Field work: Dr. Snow personally visited every home in which a cholera death occurred around his home and established their water source



London, Mid-1800s—2

- ▶ 1849: British Parliament establishes that water suppliers must move their water sources outside of central London
 - ▶ People had been complaining about the smell and residue in their water
- ▶ Two water companies:
 - ▶ Lambeth: complied with the act and drew water from upstream of the city
 - ▶ Southwark and Vauxhall: continued to draw water from central London
- ▶ Impossible to tell from a map which company supplied which neighborhood—adjacent areas could have different companies

John Snow's Grand Experiments

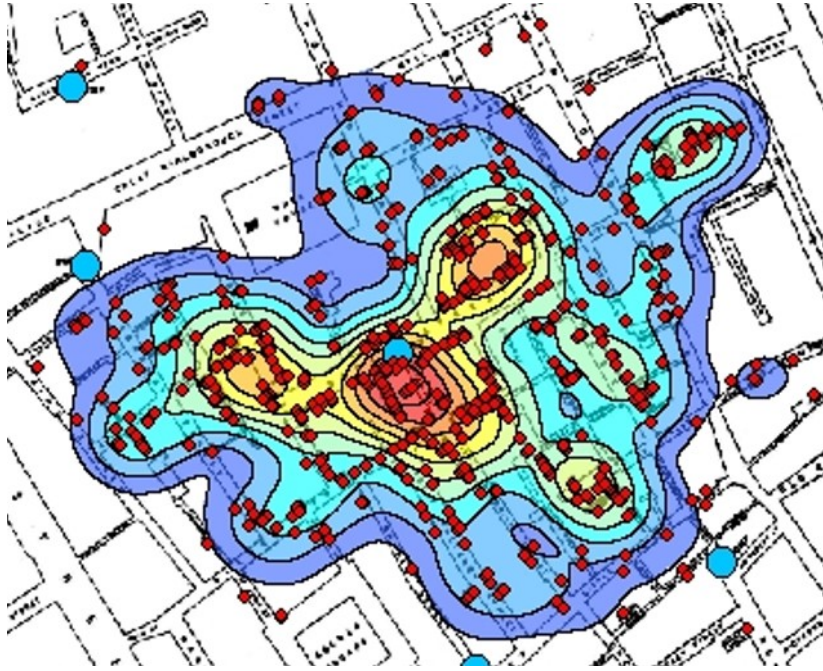


TABLE IX.

	Number of houses.	Deaths from Cholera.	Deaths in each 10,000 houses.
Southwark and Vauxhall Company	40,046	1,263	315
Lambeth Company	26,107	98	37
Rest of London	256,423	1,422	59

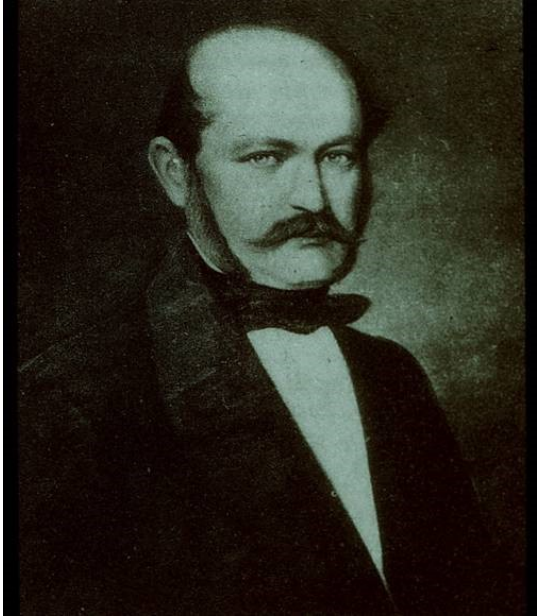
General Hospital of Vienna, Early 1800s

- ▶ A premier obstetric hospital, training physicians from all over the world
- ▶ Clinic 1:
 - ▶ Male obstetricians and medical students
 - ▶ Performed deliveries
 - ▶ Autopsied the women who died during labor
- ▶ Clinic 2:
 - ▶ Midwives, who also performed deliveries
 - ▶ Clinic assignment based on day of admission (not on case severity or maternal illness)

What Was the Problem?

- ▶ Mortality rates were substantially higher in Clinic 1 (staffed by physicians)
- ▶ In 1846:
 - ▶ Clinic 1 mortality: 11.4%
 - ▶ Clinic 2 mortality: 2.7%
- ▶ Did people notice the difference?
 - ▶ Hospital administration: ventilation and sunlight (to reduce cosmic and environmental influences) did not help
 - ▶ General public: women would delay hospital arrival or give birth in the field to try to be admitted to the midwives' clinic

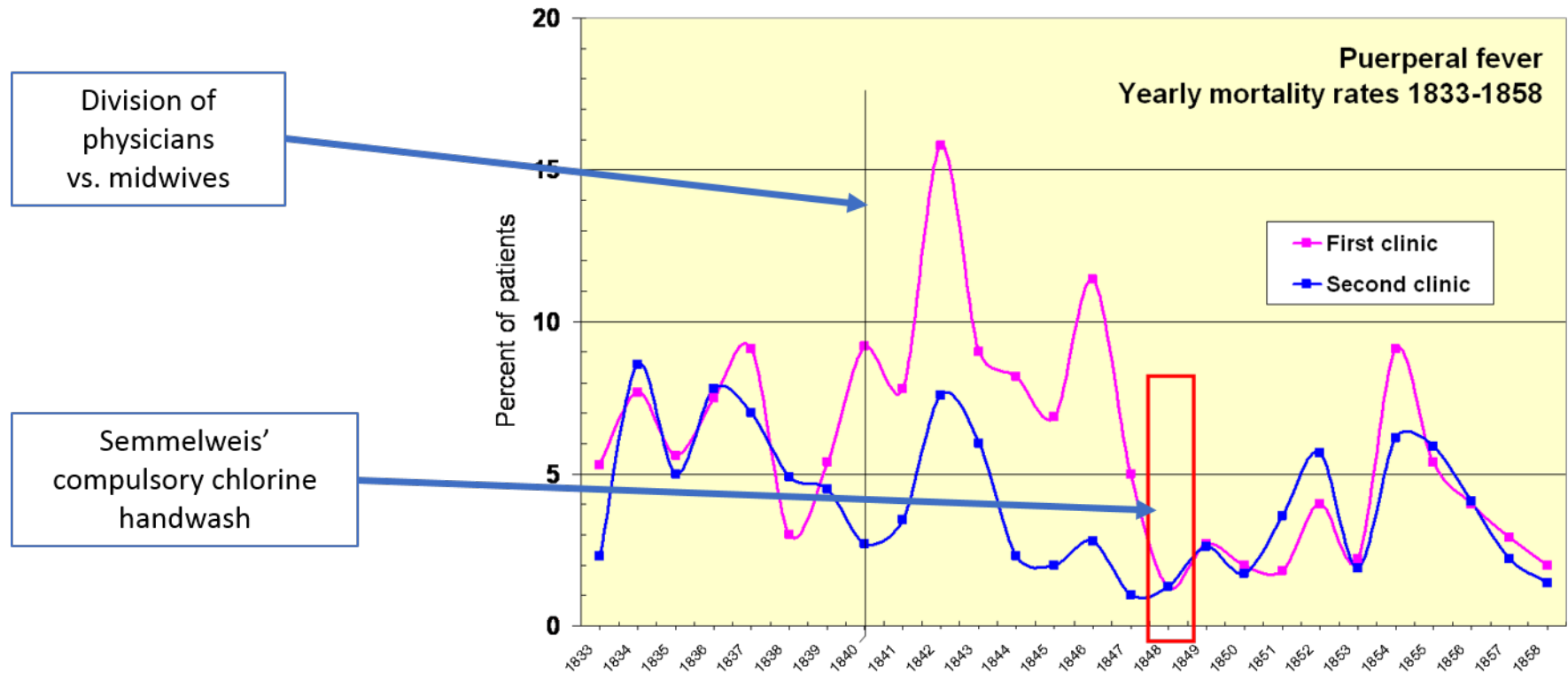
Semmelweis's Contribution



Ignaz Philipp Semmelweis

- ▶ A pathologist cut himself during an autopsy and died of an illness resembling puerperal fever
- ▶ Doctors did wash their hands with soap after performing autopsies...but their hands still smelled
- ▶ A chlorinated lime wash eliminated the smell
- ▶ May 1847: Semmelweis forced his interns to wash their hands with chlorine after autopsies

Diagnosis of a Problem and Implementation of a Solution



Are Public Health Workers Equipped to Do This?

- ▶ Preview: exercise for this course
- ▶ Think about a public health organization that you either work in or are familiar with
- ▶ Would staff working in that organization be able to conduct an epidemiologic inquiry similar to what Snow or Semmelweis did?
- ▶ Take a moment to consider this, and we will come back in the next section to talk more about what public health practitioners are expected to do and where epidemiology fits in



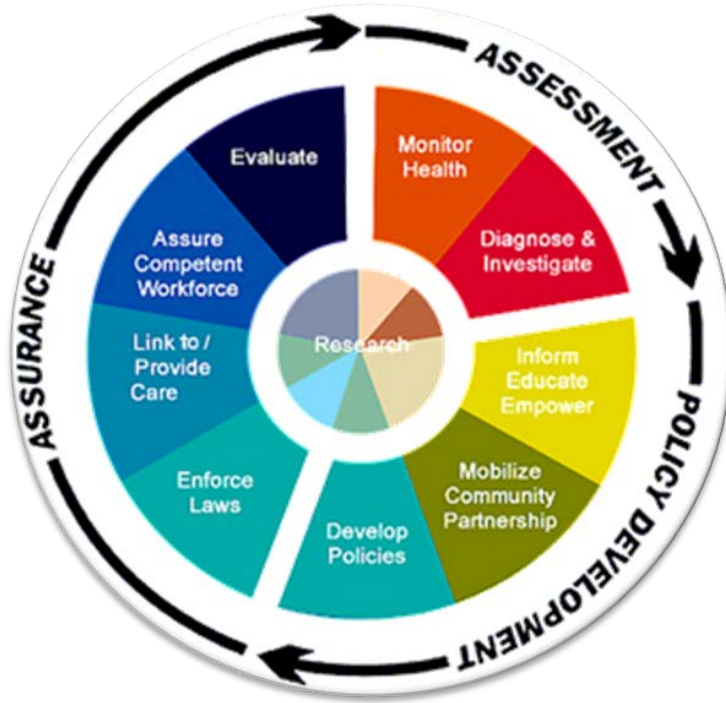
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Essential Public Health Services

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Ten Essential Public Health Services



1. Monitor health
2. Diagnose and investigate
3. Inform, educate, empower
4. Mobilize community partnership
5. Develop policies
6. Enforce laws
7. Link to and provide care
8. Assure a competent workforce
9. Evaluate
10. Research

Public Health Foundation

- ▶ Aim: improve public health and population health practice to support healthier communities
- ▶ Focus areas:
 - ▶ Performance management
 - ▶ Quality improvement
 - ▶ Workforce development

Exercise for this Course: Introduction

- ▶ Think back to your public health foundation of interest
- ▶ Recall the three tiers (or levels) of staff in a public health organization:
 - ▶ Tier 1: frontline/entry-level staff
 - ▶ Tier 2: program management/supervisory staff
 - ▶ Tier 3: senior management/executive-level staff
- ▶ The public health foundation provides a competency assessment tool for individuals to self-assess one's own competencies in each of eight skills (or domains)
- ▶ Determine what tier you do (or hope to) fit into

Exercise for this Module: Complete Competency Assessment

- ▶ On the next slide, there are links to the competency tool by tier of responsibility
- ▶ Select the appropriate link and take some time to complete a self-assessment of your ability to provide the essential public health services
- ▶ Notice in particular those competencies related to data and information or epidemiology

Exercise for this Module: Competency Tool

- ▶ Competency tool links are also available on the CoursePlus lecture page
- ▶ Tier 1: [Competency Assessment Tier 1 Public Health Professionals \(2014 version\)](#)
- ▶ Tier 2: [Competency Assessment Tier 2 Public Health Professionals \(2014 version\)](#)
- ▶ Tier 3: [Competency Assessment Tier 3 Public Health Professionals \(2014 version\)](#)



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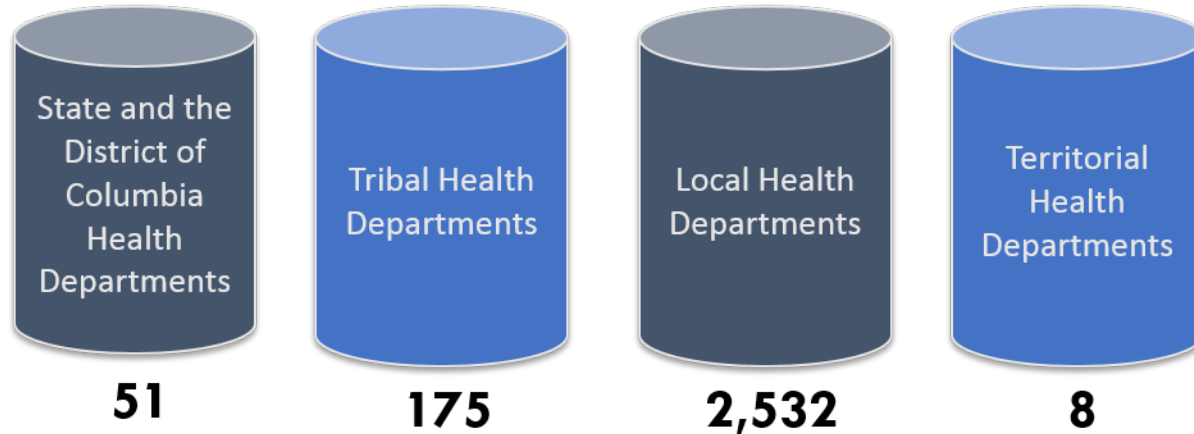
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Public Health Governance

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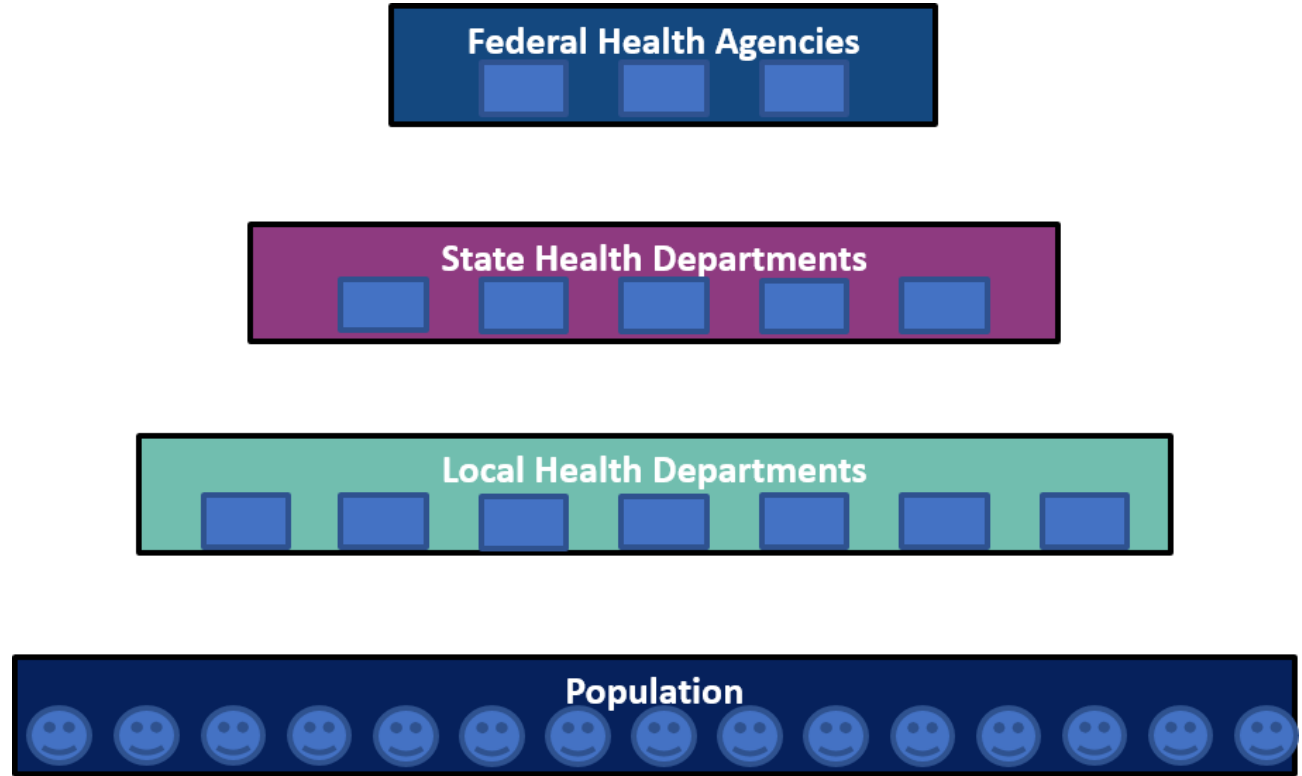
Local Public Health Governance: United States

State and local health departments
retain the primary responsibility for health under the US Constitution

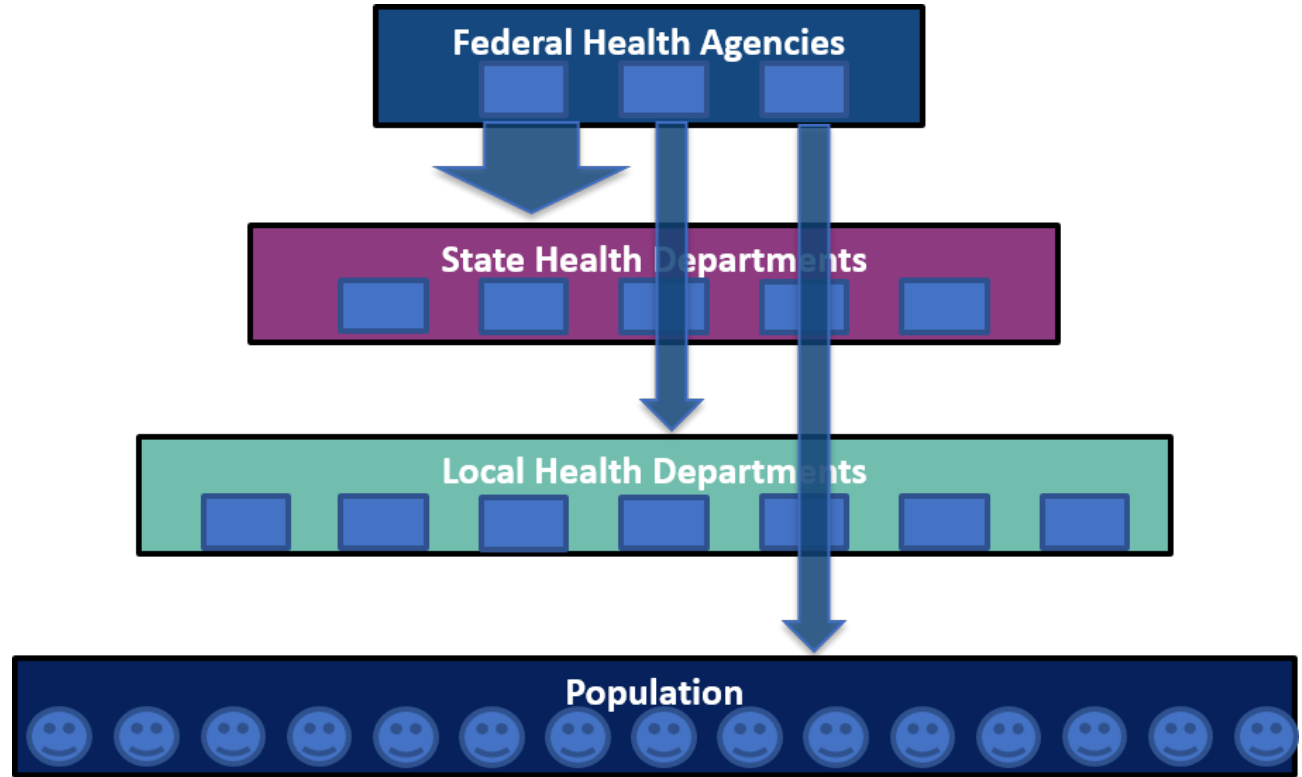


Source: National Association of County and City Health Officials (NACCHO). (2014). *2013 national profile of local health departments*. Association of State and Territorial Health Officials (ASTHO). (2011). *ASTHO profile of state public health, volume two*. National Opinion Research Center at the University of Chicago. (2012). *Final report: A profile of tribal health departments*. Retrieved October 16, 2018.

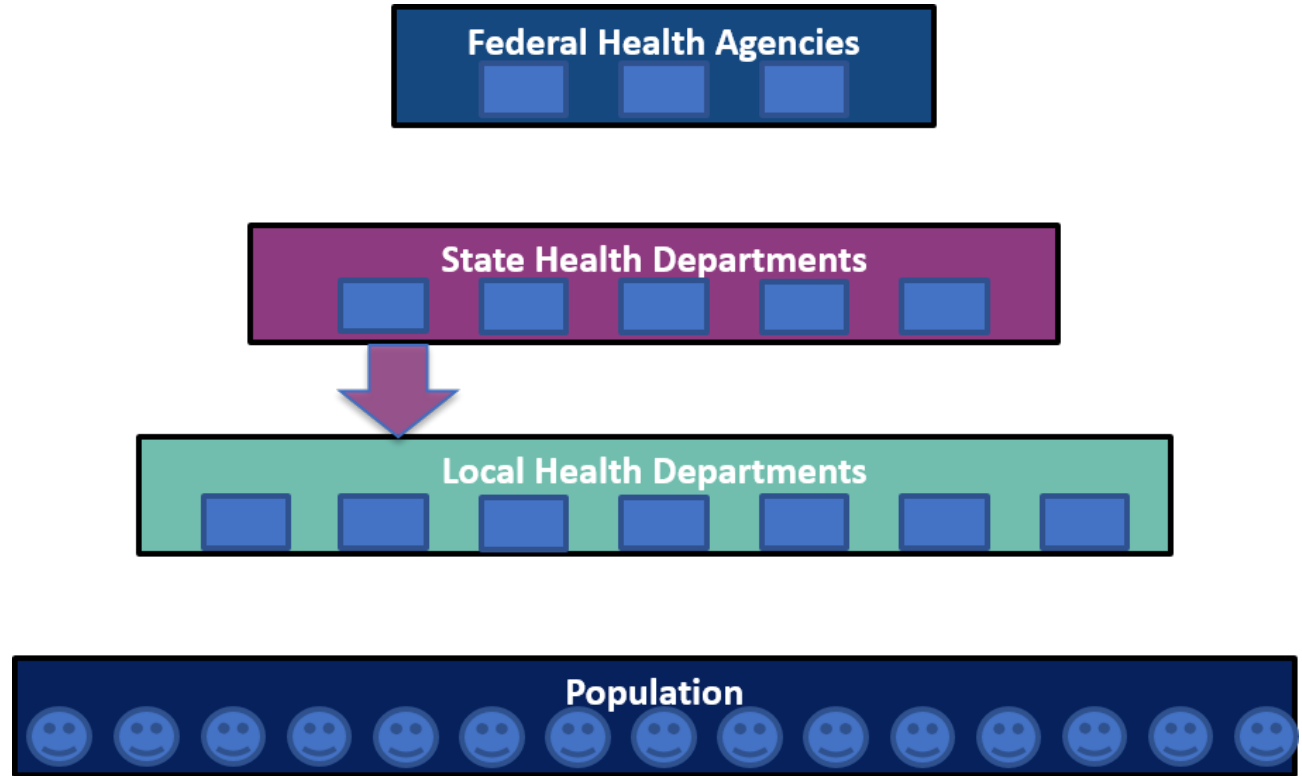
United States Public Health Infrastructure—1



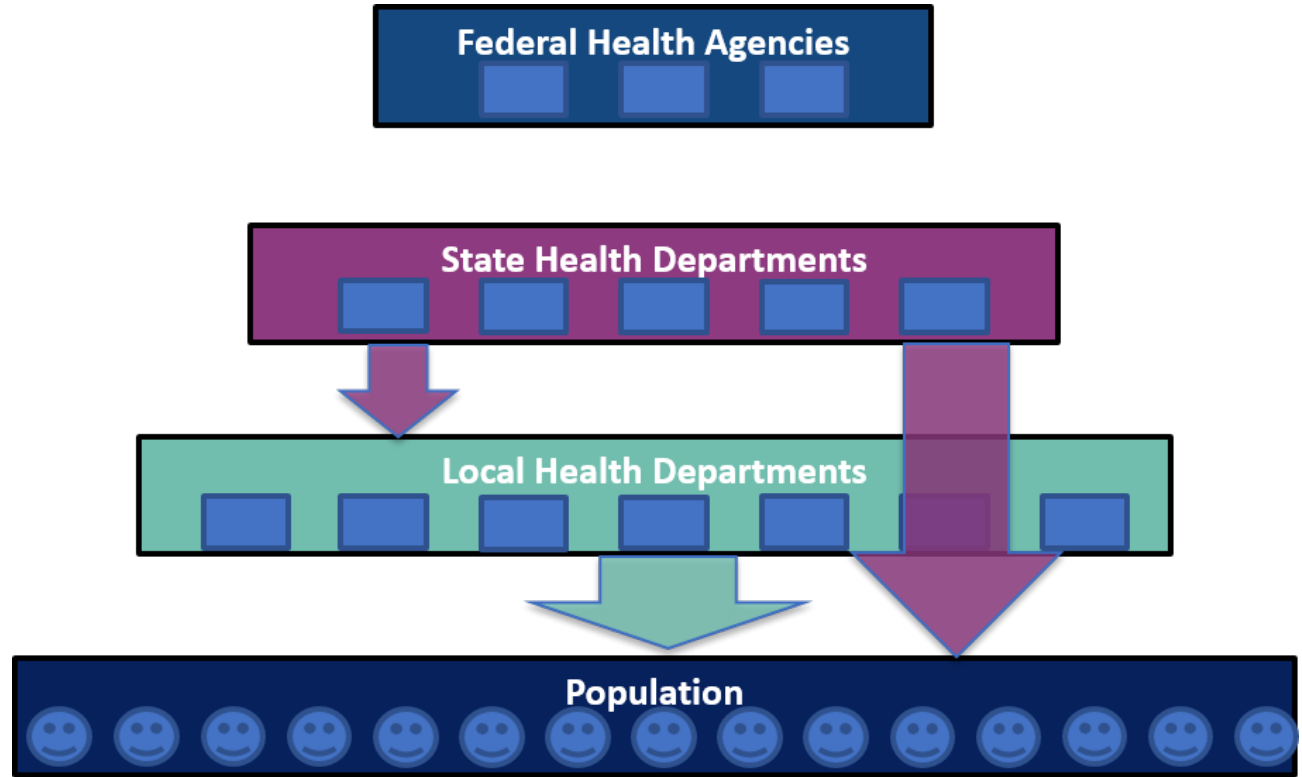
United States Public Health Infrastructure—2



United States Public Health Infrastructure—3



United States Public Health Infrastructure—4



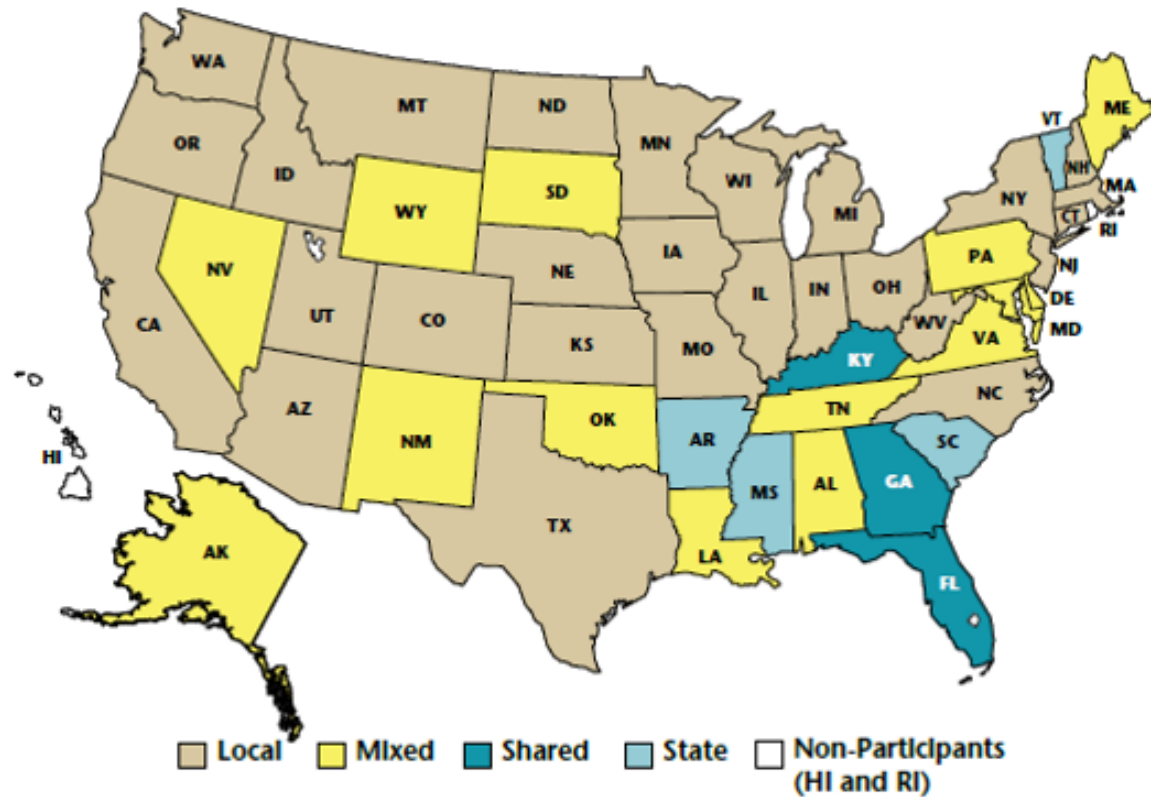
Core Functions at Government Levels

	Assessment	Policy Development	Assurance
Federal	National tobacco public health surveillance	Smoking ban on commercial flights	Federal grants for antismoking research
State	Monitor state tobacco use	Increase tobacco tax	Funding for campaign through Proposition 99
Local	Report on local tobacco use	County laws prohibiting smoking in bars	Resources to help smokers quit in multiple languages

Source: Centers for Disease Control and Prevention, Division of Scientific Education and Professional Development. (2016). Public Health 101 Series.

Local Health Department Governance: US

FIGURE 2.4 Governance of LHDs (by State)



Local = All LHDs in state are units of local government; *State* = All LHDs in state are units of state government; *Shared* = All LHDs in state governed by both state and local authorities; *Mixed* = LHDs in state have more than one governance type



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Reflection on Competency Assessment Tool Exercise

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Eight Domains

- ▶ Analytical/assessment
- ▶ Policy development/program planning
- ▶ Communication
- ▶ Cultural competency
- ▶ Community dimensions of practice
- ▶ Public health science
- ▶ Financial planning/management
- ▶ Leadership and systems thinking

Exercise for this Module: Reflection

- ▶ Review your competency assessment
 - ▶ How many of the competencies overall were you comfortable saying you could accomplish?
 - ▶ How many of the data-specific competencies were you comfortable saying you could accomplish?
- ▶ Were there particular domains that you were more or less comfortable with?
- ▶ Now think about public health organizations in general
 - ▶ Do you think our public health workforce in the US is prepared to address these competencies?

Example from the Literature

- ▶ *Journal of Community Health*, December 2003
- ▶ Dr. R. Todd Bartee and colleagues did an assessment of self-reported comfort with the recommended public health competencies among state and local health department workers in a frontier state
- ▶ Overall, the domains of communications and cultural competency were rated higher, while financial planning/management and policy development/program planning were rated lower
- ▶ The authors' recommendation: need for more targeted training of the US public health workforce in the core competencies

Where Are We Now?

- ▶ Sellers et al., 2015: survey of state health professionals showed significant needs for training in influencing policy development, preparing a budget, and training related to the social determinants of health
- ▶ Kaufman et al., 2014: series of interviews with national leadership in public health showed a need for prioritizing systems thinking, communicating persuasively, change management, information and analytics, problem solving, and working with diverse populations
- ▶ Underlying message: epidemiology, the use and interpretation of data to assess and improve the health of a population, is a core skill

Source: Sellers, K., et al. (2015). The public health workforce interests and needs survey: The first national survey of state health agency employees. *J Public Health Manag Pract*, 21(Suppl 6): S13–S27. doi: [10.1097/PHH.0000000000000331](https://doi.org/10.1097/PHH.0000000000000331); Kaufman, N. J., et al. (2014). Thinking beyond the silos: Emerging priorities in workforce development for state and local government public health agencies. *J Public Health Manag Pract*, 20(6), 557–565. doi: [10.1097/PHH.0000000000000076](https://doi.org/10.1097/PHH.0000000000000076)

Summary

- ▶ Epidemiology is the **study** of the **distribution** and **determinants of health-related states or events** in **specified populations**, and the **application** of this study to the control of health problems
- ▶ It is expected that every US resident will have access to the 10 essential public health services through local, state, and federal health authorities
- ▶ The Public Health Foundation's Competency Assessment Tools assess individuals at different levels of the public health workforce to achieve the 10 essential public health services

Thank you!